

MyStudySpace

Platform Documentation

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Date	January 2026
Domain	mystudyspace.me
Status	Production
Classification	CONFIDENTIAL

CONFIDENTIAL

Internal Use Only - Do Not Distribute

System Architecture Overview

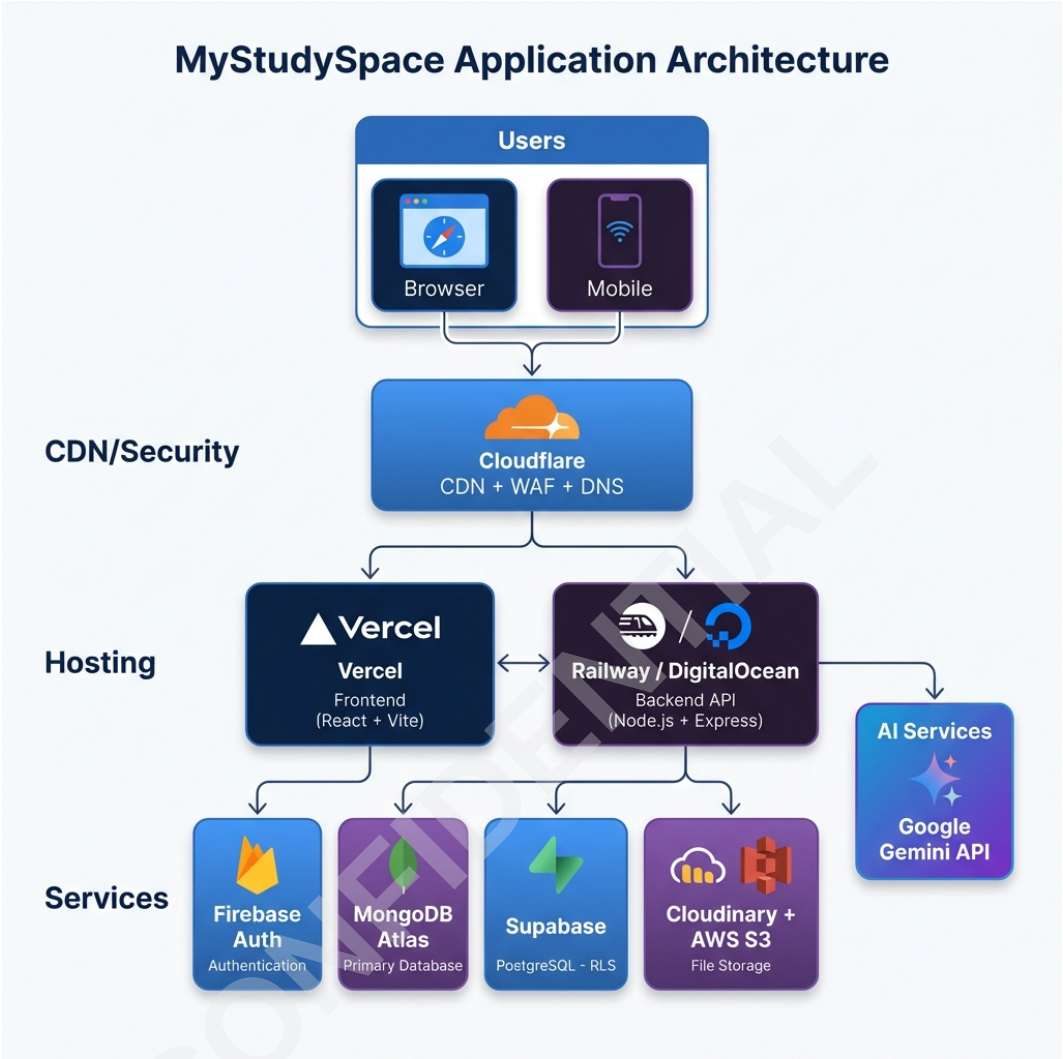


Figure: MyStudySpace System Architecture

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Executive Summary

Platform Overview

MyStudySpace is a community-driven academic resource sharing platform designed for college students to access, share, and collaborate on study materials. The platform creates a centralized ecosystem for educational content with AI-powered assistance.

Mission

To democratize access to quality study materials by enabling peer-to-peer resource sharing within verified college communities.

Key Metrics

Metric	Value
Target Institution	Krishna Institute of Engineering and Technology (KIET)
Supported Semesters	1-8
Resource Types	Notes, PYQs, Video Links
Access Modes	Full (verified email), Read-Only (guest)

Problem Statement

Challenges Addressed

Challenge	Solution
Scattered Resources	Centralized repository with organized categorization
Quality Uncertainty	Community voting system for content validation
Community Isolation	Chat rooms and threaded comments for collaboration
Information Overload	Smart filtering by semester, branch, subject
Lack of AI Support	Integrated AI assistant for study help

Product Features

Core Features

Feature	Description	Status
Resource Library	Upload, browse, and download PDFs, notes, and videos	Live
Smart Filtering	Filter by semester, branch, subject, chapter	Live
Community Voting	Upvote/downvote resources for quality surfacing	Live
Bookmarking	Save resources for quick access	Live
Following System	Follow content creators and get updates	Live

Communication Features

Feature	Description	Status
Chat Rooms	Topic-based discussion rooms (public/private)	Live
Threaded Comments	Reddit-style nested comments on posts	Live
College Notices	Official announcement system	Live
Notifications	Real-time alerts for follows, replies, mentions	Live

AI Features

Feature	Description	Status
AI Study Assistant	Chat with AI about course materials	Live
PDF-Aware Answers	AI reads uploaded PDFs to answer questions	Live
Question Generation	Generate practice questions from study materials	Live

User Roles & Permissions

Role Hierarchy

Role	Email Requirement	Permissions
Administrator	Manual assignment	Full access, user moderation, content management
Verified User	@kiet.edu domain	Upload, comment, vote, create rooms
Read-Only User	Any email	View resources, download, browse only

Permission Matrix

Action	Read-Only	Verified	Admin
View resources	Yes	Yes	Yes
Download resources	Yes	Yes	Yes
Upload resources	No	Yes	Yes
Comment on posts	No	Yes	Yes
Vote on content	No	Yes	Yes
Create chat rooms	No	Yes	Yes
Ban users	No	No	Yes
Delete any content	No	Own only	Yes

System Architecture

Architecture Overview

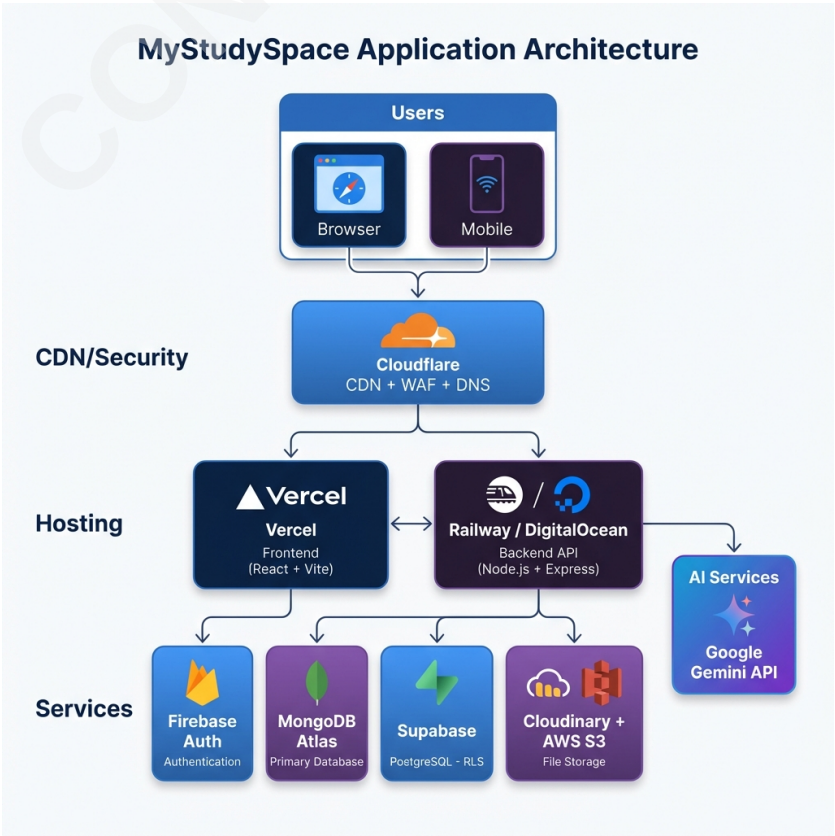


Figure: System Architecture

Infrastructure Components

Layer	Service	Purpose
CDN/Security	Cloudflare	DNS, WAF, DDoS protection, SSL
Frontend Hosting	Vercel	Edge deployment, React SPA
Backend Hosting	Railway / DigitalOcean	Node.js API server
Authentication	Firebase Auth	Google OAuth, Email/Password
Primary Database	MongoDB Atlas	NoSQL document storage
Legacy Database	Supabase (PostgreSQL)	RLS policies, transitioning
File Storage	Cloudinary + AWS S3	PDF hosting, CDN delivery, backup
AI Services	Google Gemini API	Chat assistant, question generation

Technology Stack

Frontend

Technology	Purpose
React 18	UI framework with hooks and concurrent features
Vite	Build tool and dev server
Tailwind CSS	Utility-first styling
shadcn/ui	Component library
React Query	Server state management with caching
React Router	Client-side routing

Backend

Technology	Purpose
Node.js	Runtime environment
Express.js	REST API framework
TypeScript	Type-safe development
Firebase Admin SDK	Token verification
pdf-parse	PDF text extraction for AI
Mongoose	MongoDB ODM

DevOps

Service	Purpose
Vercel	Frontend CI/CD and hosting
Railway	Backend deployment (primary)
DigitalOcean	Backend deployment (backup)
GitHub	Version control

Data Flow

Authentication Flow

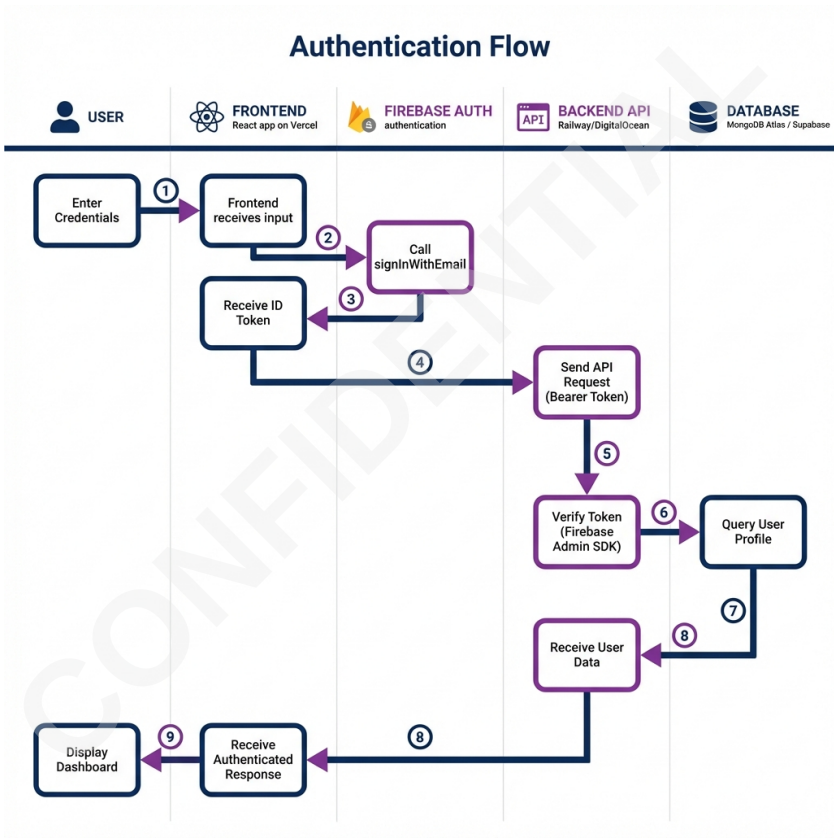


Figure: Authentication Flow

- 1. User enters credentials on frontend
- 2. Frontend calls Firebase signInWithEmail
- 3. Firebase returns ID Token
- 4. Frontend sends API request with Bearer token
- 5. Backend verifies token with Firebase Admin SDK
- 6. Backend queries user profile from database
- 7. Database returns user data
- 8. Backend returns authenticated response
- 9. Frontend displays dashboard

Resource Upload Flow

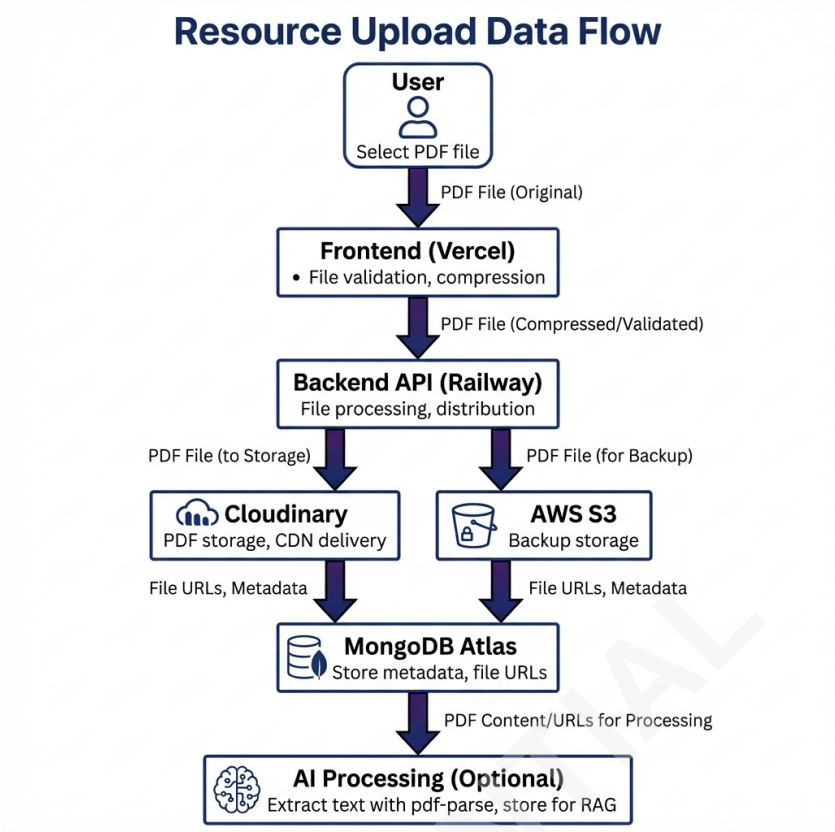


Figure: Resource Upload Data Flow

- 1. User selects PDF file
- 2. Frontend validates file type and size
- 3. Backend receives and processes file
- 4. File uploaded to Cloudinary (primary) and AWS S3 (backup)
- 5. Metadata stored in MongoDB Atlas
- 6. Optional: Text extracted for AI RAG pipeline

User Journey

Complete User Flow

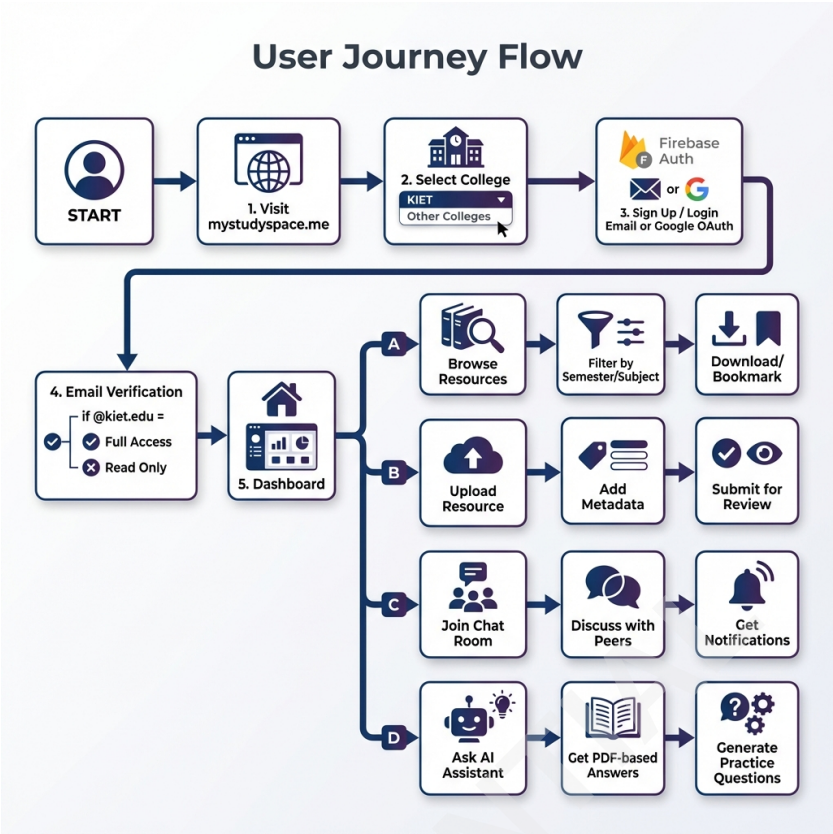


Figure: User Journey Flow

Registration Flow

1. User visits mystudyspace.me
2. Selects college from dropdown
3. Signs up with email or Google OAuth
4. Email verification sent
5. Access level determined:
 - @kiet.edu email → Full Access
 - Other email → Read-Only Access

Core User Paths

Path	Steps
Browse Resources	Filter → Search → View → Download/Bookmark
Upload Resource	Select type → Add metadata → Upload file → Submit
Join Chat Room	Browse rooms → Enter code → Join → Discuss
Use AI Assistant	Open chat → Ask question → Get PDF-based answer

Database Design

MongoDB Collections

Collection	Purpose	Key Fields
users	User profiles	email, displayName, collegeId, role
resources	Study materials	title, type, semester, branch, subject, fileUrl
chat_rooms	Discussion rooms	name, isPrivate, joinCode, createdBy
chat_messages	Room messages	roomId, authorEmail, content, parentId
notifications	User alerts	userEmail, type, title, isRead
bookmarks	Saved resources	userEmail, resourceId
follows	Social connections	followerEmail, followingEmail, status

Supabase Tables (Legacy)

Table	Purpose	RLS
resources	Resources with voting	Yes
notices	College announcements	Yes
syllabi	Course syllabi	Yes
banned_users	User moderation	Yes

Security Architecture

Security Layers

Layer	Implementation
Transport	HTTPS everywhere (Cloudflare SSL)
CDN	Cloudflare WAF, DDoS protection
Authentication	Firebase Auth (JWT tokens)
Authorization	Role-based access, server-side verification
Database	MongoDB Atlas encryption, Supabase RLS
Bot Protection	reCAPTCHA v3 on sensitive actions
Rate Limiting	IP-based throttling per endpoint

Rate Limits

Endpoint	Limit	Window
Authentication	10 requests	15 minutes
AI Chat	5 requests	1 minute
File Upload	20 requests	1 hour
General API	100 requests	1 minute

Performance & Scalability

Optimization Strategies

Strategy	Implementation
Code Splitting	Route-based lazy loading with React.lazy()
CDN Caching	Cloudflare edge caching for static assets
Image Optimization	Cloudinary transformations, WebP format
API Response Caching	React Query with 5-minute stale time
Database Indexing	Compound indexes on frequently queried fields

Scalability

Component	Scaling Strategy
Frontend	Vercel auto-scaling edge functions
Backend	Railway horizontal scaling
Database	MongoDB Atlas auto-sharding
Files	Cloudinary + AWS S3 global CDN

Privacy & Compliance

Data Collection

Data Type	Purpose	Retention
Email address	Authentication, notifications	Until account deletion
Profile info	Display, personalization	Until account deletion
Upload activity	Attribution, moderation	Permanent
Usage analytics	Performance monitoring	90 days

User Rights

- Right to access personal data
- Right to data deletion (account removal)
- Right to data portability
- Opt-out of non-essential communications

Future Roadmap

Phase 2: Product Enhancement (Q2 2026)

Feature	Description	Status
Android App	Native Android app with Jetpack Compose, Material 3	In Development
AI Content Summarization	Automatically create concise summaries of PDF documents	Planned
Intelligent Doubt Resolution	AI suggests relevant resources based on student queries	Planned
Admin Bulk Actions	Bulk approve/reject resources for administrators	Planned

Phase 3: AI Premium Features (Q3 2026)

Feature	Description	Priority
Auto Question Generation	AI analyzes notes to generate practice questions	High
Plagiarism Detection	Check uploaded content for originality	Medium
Auto-Tagging	Automatically categorize and tag resources	Medium
Personalized Recommendations	ML-based content suggestions from study patterns	High

Phase 4: Platform Expansion (Q4 2026)

Milestone	Target
Multi-College Federation	10+ colleges in Delhi-NCR
LMS Integration	Moodle, Canvas compatibility
Advanced Analytics	Engagement dashboards for colleges
B2B Sales Launch	First 3 paying institutional customers

Long-Term Vision (2027+)

- Pan-India expansion to 50+ engineering colleges
- 100,000+ active students on platform
- Revenue target: █10-12 lakhs ARR
- Native iOS app development
- Video streaming and live study sessions

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